



Facilitator Guide

Sector
Green Jobs

Sub Sector
Waste Management

Occupation
Aggregator

Reference Id
SGJ/Q6201, Version 2.0 NSQF Level 3



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Agri-Residue Aggregator

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“

Skilling is building a better India.
If we have to move India towards
development then Skill Development
should be our mission.

”

Shri Narendra Modi
Prime Minister of India





Certificate

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Complying to National Occupational Standards of
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About this Guide

The objective of this book is to provide theoretical and practical training to become a Safai Karamchari. The SGJ /Q6201- Agri Residue Aggregator Pack includes the following national commercial standards (six):gSa%

1. SGJ/N6201 - Assessment of demand and supply of agro-residue
2. SGJ/N6202 - Purchase of agro-residue stock from nodal point
3. SGJ/N6203 - Packing and Storing Compacted Agricultural Residues
4. SGJ/N6204 - Sale and transportation of agricultural residues
5. SGJ/N6205 - Work effectively in a team and maintain safety

Symbols Used



Ask



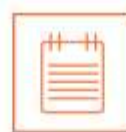
Demonstrate



Facilitation Notes



Learning Outcomes



Notes



Objectives



Practical



Team Activity



Do



Explain



Say



Resources



Activity



Summary



Role Play



Example

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Employability Skills

It is recommended that all trainings include the appropriate Employability skills Module.



Content for the same can be accessed

Employability skills workbook







1. Introduction

Unit 1.1 - Farm Waste and its Types

Unit 1.2 - Role of Agricultural Residue Aggregator



Key Learning Outcomes



At the end of this module, you will be able to:

1. Identifying farm-waste and its sources
2. To identify the types of farm-waste
3. To identify the market for farm-waste
4. To know the benefits of managing agricultural residues
5. To know the importance of work of agricultural-residue aggregator
6. To know the steps of biomass supply chain management
7. Identify the work and responsibilities of an agro-residue aggregator
8. To know the tools and equipment used by the agricultural-residue aggregator

Unit 1.1 - Farm Waste and its Types

Unit Objectives

At the end of this unit, you will be able to:

1. Identifying farm-waste and its sources
2. To identify the types of farm-waste
3. To identify the market for farm-waste
4. Know the benefits of managing agricultural residues

Facilitation Notes

- Tell students about the different types of waste that a farm produces and give some examples of farm waste. Also point out that nothing is really a waste on the farm because it can be used in some way or the other.
- In agriculture and farming operations, animals, seeds, machines, fertilizers are used. It creates different types of agricultural wastes. This agricultural waste can be separated into agricultural waste and animal waste.
- Agricultural waste is further divided into two parts: the first is the waste that comes out in processing and the second is the crop residue.

Explain

- Show them some pictures of crop residues.
- Tell them that processing waste is divided into primary waste and secondary waste.
- The primary residue comes out while cutting the crops. Examples of such wastes are rice straw, sugarcane tops, maize stalks, etc.
- A waste of residue, released during the processing of crops. Examples of such residues are coconut, rice husk etc.
- Tell them about animal waste and its uses. In India, animal waste has its own importance.
- In ancient India, it had many uses such as use as cooking fuel, fertilizer to increase soil fertility, etc. But due to increase in pollution and health concerns due to its use as a cooking fuel, the government has restricted its use.
- and is encouraging people to look at other available options.
- Explain the process of making biogas to the students.
- Discuss with them the benefits of biogas.
- For safe and productive use of animal waste, farmers have started making biogas from it.
- It is safe, pollution free, cheap and a better option as a renewable source of energy.
- Explain the sources of agricultural residues to the learners.
- Inform them that India has about 200-240 million tonnes of surplus residues out of a total of 600 tonnes million residues. This relic comes in different forms from different states. Tell them about the major rice and wheat producing states of India.
- Educate the learners about the benefits of proper management of agri residues.
- Tell them:
 - It is necessary to keep the environment safe and clean.
 - It is really helpful for making biomanure which is further beneficial for increasing the fertility of the soil.
 - It can be good for making biogas which will further power generation which will be used in transportation mediums and as cooking fuel.

Ask



- For example, most of the rice husk is produced from rice producing states like West Bengal, UP etc.
- Most of the wheat straw comes from states like UP, Haryana and Punjab.

Unit 1.2 - Role of Agricultural Residue Aggregator

Unit Objectives

At the end of this unit, you will be able to:

1. To know the importance of work of agri-residue aggregator
2. To know the steps of biomass supply chain management
3. Identify the work and responsibilities of an agro-residue aggregator
4. To know the tools and equipment used by the agricultural-residue aggregator

Explain

- Explain the basic duties of an Agricultural Residue Aggregator to the learners. Ask them to fulfill their duties with sincerity and hard work.
- Tell them that in biomass supply chain management, different types of agricultural produce waste related to harvesting, processing, storage and transportation are collected and then processed to make biofuels.
- Brief them about each step of supply chain management.

Facilitation Notes

- Help them solve their queries and share some tips to complete their work on time.
- Guide them on managing the biomass supply chain.

Ask

- Ask them about their doubts and issues that they face while working
- Ask them about their doubts.

Do

- Display some images of the equipment used by the Agricultural Residue Aggregator.
- Briefly explain the uses and advantages of each tool.
- Discuss with them the roles and responsibilities of an agricultural residue aggregator.



2. Assessment of demand and supply of agro-residue



Unit 2.1 - Estimating Demand and Supply of Agro-residue

Unit 2.2 – Identifying Supplier for Planning Supply of Agro-Residue Stock



Key Learning Outcomes



At the end of this module, you will be able to:

1. To know the status of present agro-residue market
2. To know the trend of agricultural residues in the market
3. Identifying and Contacting Suppliers/Farmers
4. Negotiating with the buyer
5. Fixing the Price and Placing the Order
6. Knowing the steps to maintain stock
7. Agricultural – To find out the right quality of residues

Unit 2.1 - Estimating Demand and Supply of Agro - residue

Unit Objectives

At the end of this unit, you will be able to:

1. To know the status of present agro-residue market
2. To know the trend of agricultural residues in the market.

Explain

- Inform and guide learners on how to accurately predict market trends.
- Tell them that every region is different and hence there are different crops, so there should be information about the weather, crops and its share of a region in the total residual market.

Ask

- Ask them to estimate the buyers demand accurately and timely to avoid any last minute hassles.
- Ask them to estimate the quantity of the total residue and the reasons for the demand for the residue by the buyers. This will help them to capture the market in a proper way.

Unit 2.2 – Identifying Supplier for Planning Supply of Agro-Residue Stock

Unit Objectives

At the end of this unit, you will be able to:

1. Identifying and Contacting Suppliers/Farmers
2. Negotiating with the buyer
3. Fixing the Price and Placing the Order
4. Knowing the steps to maintain stock
5. Finding the Right Quality of Agricultural Residue

Explain

- Tell them that farmers are the main suppliers of agri-residue and it is important to establish a good rapport with them to maintain the balance between demand and supply in the market.
- Tell them the main reasons for making good relations with the farmers.
- Help them prepare a sample sales order.
- Tell them what details should be mentioned in the order of sale.
- Tell them that it is important to check the quality of the residue when buying.

Facilitation Notes

- Briefly describe the main reasons for the demand of agro-residues in the market.
- Tell them the whole process of buying agricultural residues.
- Suggest some key points to them while fixing the correct rate for the residuals.
- Tell them the importance of documenting every purchase.
- The density of the bundle of residues should not be too high as it would make transportation difficult and should not be too low as it would be difficult to handle.
- Residue bundles should not be too moist as they are not useful for long term use.

Ask

- Ask them if they face any issues while working with farmers and suggest some ways to help them.
- Tell them to ask: Will the quality and type of agricultural residues suit their needs?
- What is the quantity they need and how quickly they repeat their needs.

Say

- Say them to maintain a good relation with the buyers and understand their requirements from time to time.



3. Purchase of agro-residue stock from nodal point



Unit 3.1 - Establishing Nodal points for collection

Unit 3.2 - Collection and Receipt of Agro-Residue Stock



Key Learning Outcomes

At the end of this module, you will be able to:

1. Identifying the steps for setting up collection points for procurement of stock from farmers
2. Coordinate with suppliers to buy stock at low cost
3. Identifying tools and equipment to be used at the collection point
4. Telling the whole process of collecting/receiving agriculture-residue
5. Managing Purchased Stock

Unit 3.1 - Establishing Nodal points for collection

Unit Objectives

At the end of this unit, you will be able to:

1. Identification of area to set up collection point
2. Telling the steps for proper installation of collection point

Explain

- Tell the learners that a good collection point should be carefully determined.
- Tell the learners that after setting up a collection centre, it is important to inform the farmers about the same.
- Tell them the importance of having a regular meeting and discussion with the farmers:
- To maintain the demand and supply balance in the market
- To fix the right price for agricultural residues
- To inform them about the requirements of the market
- Describe the most common equipment used in a collection center.
- Show them some pictures of each tool and explain their uses.

Facilitation Notes

When selecting a collection point, it should:

- Make a list of future wellness centers and note down the facilities available there. The location should be planned with all the necessary equipment like machinery, equipment and sheet cover.
- The place should be safe and secure in all seasons like rainy and winter.
- This place is easily accessible to the farmers and their tractors.

Unit 3.2 - Collection and Receipt of Agro-Residue Stock

Unit Objectives

At the end of this unit, you will be able to:

1. Identifying the process of collecting agricultural residues according to quality, quantity and cost.
2. Listing the steps to buy agri-residue.
3. Management of stocks and transportation of agricultural residues.
4. Coordinating with suppliers for procurement of agri-residue
5. Identifying the tools and equipment to be provided at the collection point

Facilitation Notes

During this process discuss with them some preventive measures such as:

- Bundles of different moisture content should be kept separately otherwise it may damage some of the bundles.
- The density of each residue bundle should be in a specified range as too little or too much may restrict its easy transport.
- Maintain complete record of purchases and prices.
- Tell the learners to select the mode of transport as per the requirement and cost expenditure i.e. it should be in budget and time saving.

Explain

- Explain to the learners the whole process of collecting agro-residue from the farmers.
- Discuss some methods for easy and safe transportation of agricultural residues.

Do

- Check the quality of the residue to ensure its suitability as per the requirements.



4. Packing and Storing Compacted Agricultural Residues



Unit 4.1 - Compaction and Packing of Agricultural Residues

Unit 4.2 - Storing Agricultural Residue



Key Learning Outcomes



At the end of this module, you will be able to:

1. To grade the agro-residue and tell the method of sorting.
2. To identify the characteristics of agro-residue before making bales.
3. Telling the process of making bundles of agriculture-residue and collecting the bundles.
4. Listing the steps for packing the bales.
5. To identify the precautions and methods of storage of bales.

Unit 4.1 - Compaction and Packing of Agricultural Residues

Unit Objectives

At the end of this unit, you will be able to:

1. To grade the agro-residue and tell the methods of sorting.
2. To identify the characteristics of agro-residue before making bales.
3. To know the process of making bundle of agriculture-residue.
4. To identify the quality of bundle.

Facilitation Notes

- Add some examples to their answer.
- Tell the learners about the importance of quality checking while preparing the residue bundle.

Explain

- Introduce the learners to the grading process of agro-residues.
- Tell them that there are two types of agricultural residues.
- Walk the learners through the entire process of collecting and packing agri-residue bundles

Ask

- Ask them if they face any problem while doing this task.

Say

- Say them to give some examples of each.
- Say them to ensure about the quality, moisture, quantity and density of the bundles.

Unit 4.2 - Storing Agricultural Residue

Unit Objectives

At the end of this unit, you will be able to:

1. Telling the steps to pack the bundles.
2. To identify the precautions and methods of storing the bales.

Facilitation Notes

- Inform the learners about the importance of safe storage of farm-residue bundles.
- Tell them that it is necessary to store such stock in a building that is well ventilated, environment safe and at room temperature.



5. Sale and transportation of agricultural residues



Unit 5.1 - Marketing and Sales of Agricultural Residue

Unit 5.2 - Transporting Agricultural Residues



Key Learning Outcomes

At the end of this module, you will be able to:

1. Understand the selling position of the agri-residue market.
2. Identifying and coordinating with potential buyers for agro-residue waste.
3. Telling about how to create a sales order.
4. Maintaining relationship with the customer.
5. Identifying the correct method of transportation of agricultural residues.
6. Choosing the right transport.

Unit 5.1 - Marketing and Sales of Agricultural Residue

Unit Objectives

At the end of this unit, you will be able to:

1. Understanding the sales conditions of the agri-residue market.
2. To identify potential buyers for agro-residue waste.
3. Agriculture - Coordinating with them for residue waste.
4. Telling about how to create a sales order for residuals.
5. Maintaining relationship with customers.

Facilitation Notes

Suggest some good ways to maintain a good bonding with the customers.

Explain

- Inform the learners about each step in the sale and marketing process of agri-residues.
- Tell the learners that maintaining good relationship with customers is essential in every field of work.

Ask

- Ask them if they have any doubts or want to share their customer experience.

Say

Say three students to role play by playing the roles of farmer, aggregator and supplier to show the duties.

Unit 5.2 - Transporting Agricultural Residues

Unit Objectives



At the end of this unit, you will be able to:

1. Identifying the right way to transport agricultural residues.
2. Identifying the most suitable vehicle for transportation.
3. Explaining the measures to be taken while transporting.

Explain



- Tell the learners about the different modes of transportation of agricultural residues like animal trolley, truck, tempo etc.
- Inform the learners about the whole process of transportation of agri-residues.

Facilitation Notes



Give them some tips on how to choose the right mode of transport as per the requirement.



6. Working effectively in a team and maintaining safely



Unit 6.1 - Working Effectively in a Team

Unit 6.2 - Maintaining Health and Safety
at the Workplace



Key Learning Outcomes



At the end of this module, you will be able to:

1. Identify the right and wrong way of communication
2. List the measures to maintain good health and safety at workplace

Unit 6.1 - Working Effectively in a Team

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the types of communication required in this role
2. Describe the importance of listening
3. Define the purpose of interaction with supervisors
4. Identify the Purpose of Interacting with Colleagues
5. Recognize the right and wrong way of communication

Facilitation Notes

- Tell the learners what is an effective communication.
- An effective communication is made up of:
 - being polite and patient
 - Familiarity with native and English language
 - being able to resolve conflicts
 - being helpful and a good listener
- Explain to the learners the importance of maintaining a good communication and a regular interaction with the supervisor and co-workers. This will help them build a good image, become a reliable and respected person and a problem solver.
- Tell them about the need to have a regular communication with the supervisor:
 - To receive work instructions and report job status
 - Setting timelines and quality standards
 - Understanding and following company guidelines and instructions
 - To solve common problems
- Tell them about the importance of working and collaborating with colleagues to achieve personal and organizational objectives. Guide them about the various dos and don'ts of an effective communication at the workplace.

Ask

- Ask them whether they are aware of the different modes of communication. If not, tell them about verbal, non-verbal, verbal and written communication.
- Ask them how often they talk to their colleagues and how it is helpful to their work.

Do

- Ask them to give examples of each type of communication.
- Ask them to role play to demonstrate effective and professional communication.
- Ask them to make a sample document of agri-residue purchase as per the organizational format.

Unit 6.2 - Maintaining Health and Safety at the Workplace

Unit Objectives

At the end of this unit, you will be able to:

1. Health and Safety Standards
2. Describe the safety measures for accident prevention
3. Personal Health
4. Identification of safety signs
5. Explain the precautions to prevent fire accident
6. Identify Safety Procedures

Facilitation Notes

- Tell them the importance of keeping the workplace safe and clean.
- Inform them about the problems and how to escalate the issues to the higher authorities.
- List some unsafe acts and conditions that every employee should avoid while working.
- Tell them that it is everyone's responsibility to keep themselves and others safe in the workplace.
- Tell them that a workplace needs a health and safety policy for the following reasons:
 - To show full commitment to the health and safety of employees
 - To prove to employees that they are safe at their workplace
 - To have an overview of the company's objectives, principles, plans, ideas and procedures
 - Fix the accountability of the organization for the health and safety of the workers
 - Complying with the Occupational Health and Safety Act
 - Comply with all health policies related to any injury to an employee
- Explain the types of hazards that an inefficient manager may face and the causes of such hazards
- Tell the learners about each component of PPE such as:
 - Helmet will help to protect their head from unexpected fall.
 - The shoes will help them prevent any slippage.
 - No ring should be worn on the finger to prevent it from getting caught with the object.
 - Wear goggles in case of wiring work.
 - Hand and arm protection is required to keep one safe from fire sparks.
- Tell the learners that a manager should know to ensure safety from fire accidents.
 - To ensure fire safety, a waste management manager must attend safety workshops or fire drills organized by the organization.

- There should be “No Smoking” signboards in sensitive areas in the organization.
- He should frequently check the work place to ensure the condition of wires, circuits, switches etc.
- Tell them that a fire drill is a practice for evacuating a building in an emergency.
- Tell the learners about the importance of arranging fire drill at work place as it reduces the chances of fire accidents and keeps everyone aware about the fire guidelines and instructions
- Tell them that it is important to keep yourself safe when using a fire extinguisher.
- Tell them that different types of fire extinguishers are used for different types of fires.

Ask

- Ask them to protect themselves from such hazards by wearing PPE if necessary.

Do

- Demonstrate some common safety signs and symbols used in the workplace.
- Demonstrate them the process of operating a fire extinguisher.





Employability skills

It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed

[Employability skills workbook](#)



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Annexure



Module No.	Unit No.	Topic Name	Page no.	URL	QR Code(s)
1	Unit 1.1	Farm waste and its types	12	https://www.youtube.com/watch?v=ZhTpljfk2NE&t=31s	
2	Unit 1.2	Role of Agricultural Residue Aggregator	19	https://www.youtube.com/watch?v=B_0nLwT5ato	
3	Unit 2.1	Assessing Demand and Supply of Agricultural Residue	24	https://www.youtube.com/watch?v=5wTp_RDeCHA	
4	Unit 2.2	Identification of suppliers for planning supply of agricultural residue stock	30	https://www.youtube.com/watch?v=cuwCgvVJzwQ&t=2s	
5	Unit 3.1	Establish nodal point for collection	40	https://www.youtube.com/watch?v=2ooR1vHweco	
6	Unit 3.2	Collecting and receiving agricultural residue stock	47	https://www.youtube.com/watch?v=BUx1O27PpsQ	
7	Unit 4.1	compaction and packing of agricultural	53	https://www.youtube.com/watch?v=Nnkb1tJowZA	

		residues			
8	Unit 4.2	storing agricultural residues	57	https://www.youtube.com/watch?v=LsdV-lrOcg	
9	Unit 5.1	Marketing and Sales of Agricultural Residue	62	https://www.youtube.com/watch?v=s6JmrNNeFFQ	
10	Unit 5.2	transport agricultural residues	67	https://www.youtube.com/watch?v=RkwQpyW-ne4	
11	Unit 6.1	maintaining health and safety at work	87	https://www.youtube.com/watch?v=KG3yvfEdiJU	
12	Employability skills	personal potential and value	Nil	https://www.youtube.com/watch?v=GlTt3DAy8cg	
13		Digital Literacy Iterator	Nil	https://www.youtube.com/watch?v=4DkM3wUh0dY	
14		money matters	Nil	https://www.youtube.com/watch?v=vG-8xVhFZhQ	

					
15		Preparing for employment and self-employment	Nil	https://www.youtube.com/watch?v=RWvslwwijgc	
16		understanding entrepreneurship	Nil	https://www.youtube.com/watch?v=xgt8QuYw83I&list=PLofZcNZJ8Dg5boOfR2Plhlu3TIP0wGajC&index=15	
17		prepare to be an entrepreneur	Nil	https://www.youtube.com/watch?v=FC2qD149JrM&t=349s	





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